Technical Report: Graphic User Interface (GUI) for the MARSIS team

1. Development of the GUI

A Graphical User Interface (GUI) was developed to assist the MARSIS team during the planning stage. The goal was to deliver the prediction and optimization techniques in a way that integrates with the already established pipeline, without sacrificing human involvement. To support this process, the GUI was equipped with various graphical representations highlighting key parameters of each observation. The result is an intelligent interface capable of conveying the operational context and supporting the MARSIS team in decision-making.

2. Testing and Analysis

The GUI is divided into different pages. The “Main Page” is used for configuring the algorithm, where all operations can be performed. The “Loading Page” is displayed while waiting for the results. The “Result Page,” which is not used for retraining, is the final page and displays the output of the algorithm.

2.1 Main page

The Main Page (Figure 1) is composed of sections for identifying the operation to be performed. The “Search for a loading file” option is used to locate the file required to run the algorithms. If the neural network needs to be retrained, the “Retraining” checkbox can be selected. After that, the process is initiated by clicking on the “Upload” button.

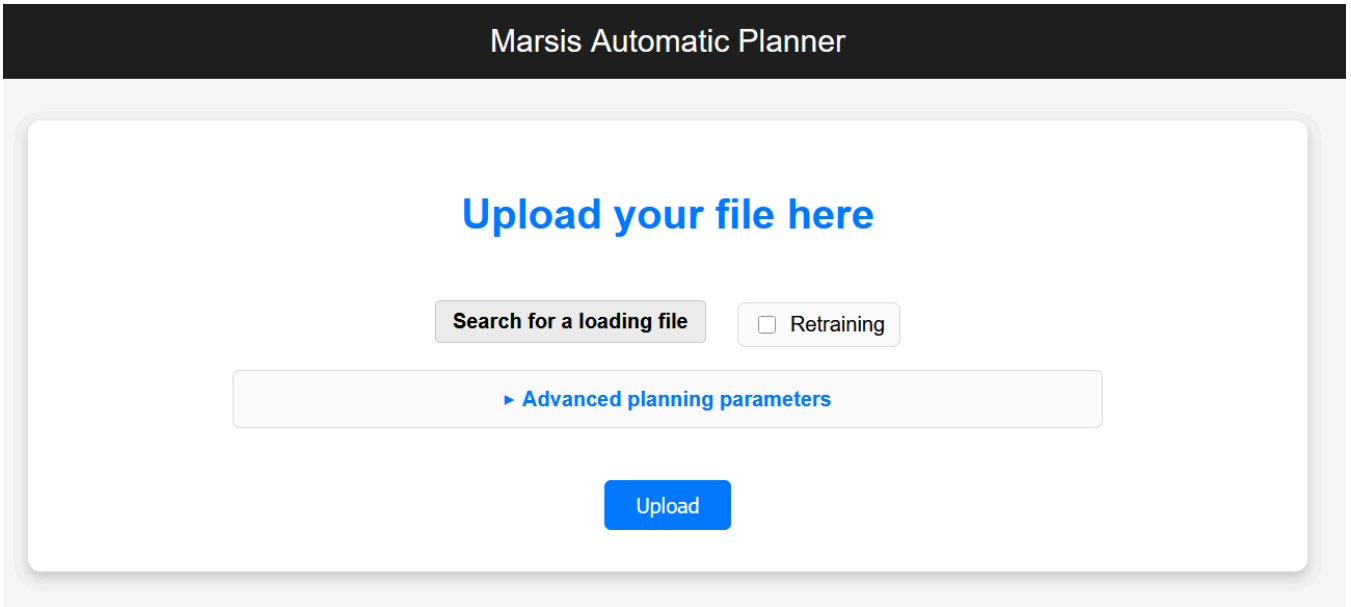

Figure 1

The “Advanced Planning Parameters” section can be expanded to modify computational variables (except for updates). The pre-selected values were defined in collaboration with the MARSIS team. The “Quality Variable” represents the target value of the neural network and optimization process, and it can be set to either “SNR” or “Peak Power.”
The computation can also include previously covered points during the optimization phase by selecting the “Consider previously covered points” option. The Sun Elevation Angle (SEA) cap value can be adjusted using the “Sun Elevation Angle Cap Value” parameter, while the operational time interval of the satellite—expressed as seconds around perihelion—can be modified through the “Time Off Peri Cap Interval.”
Additionally, the four quality classes are defined using the “Separation Quality Variable,” where the threshold values for each class are specified.

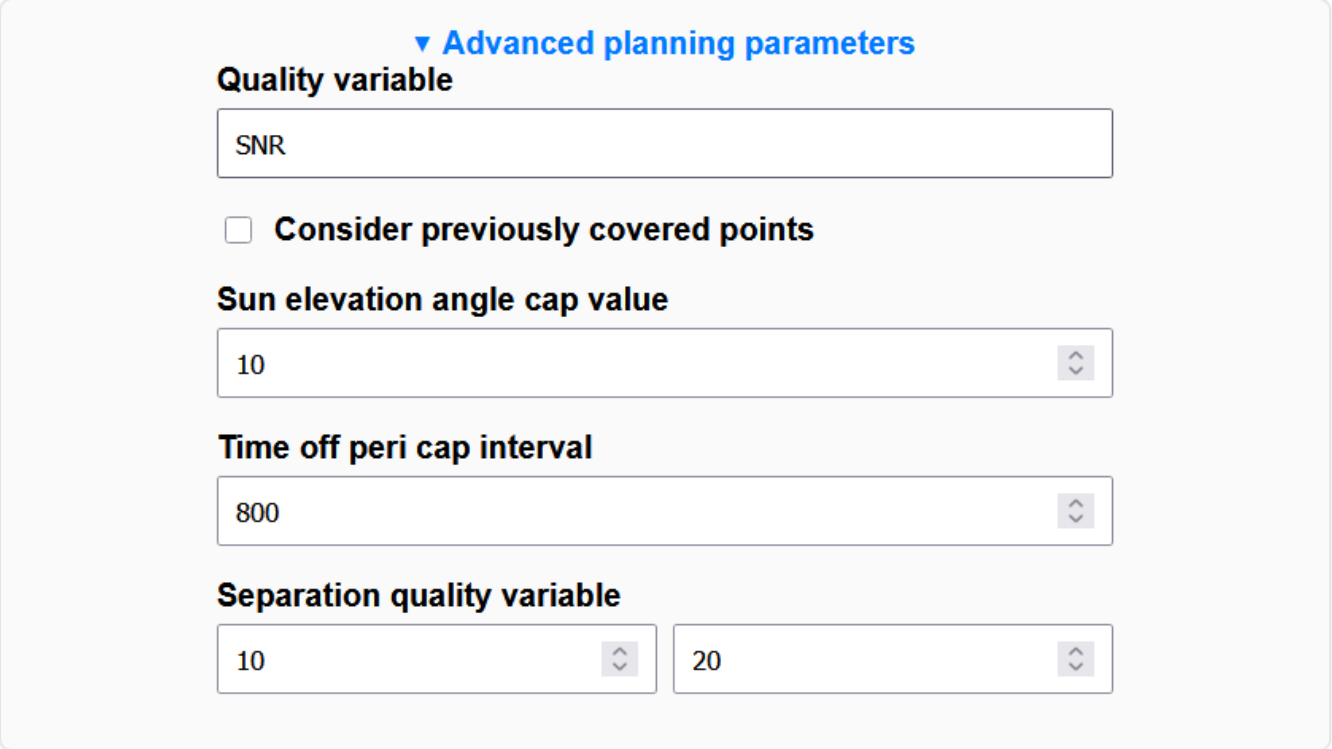

Figure 2

2.2 Loading page

The loading page (Figure 3) appears while the planning is undergoing, and it consists of a round shape which shows a repeated movement. It will disappear when the operation is done.

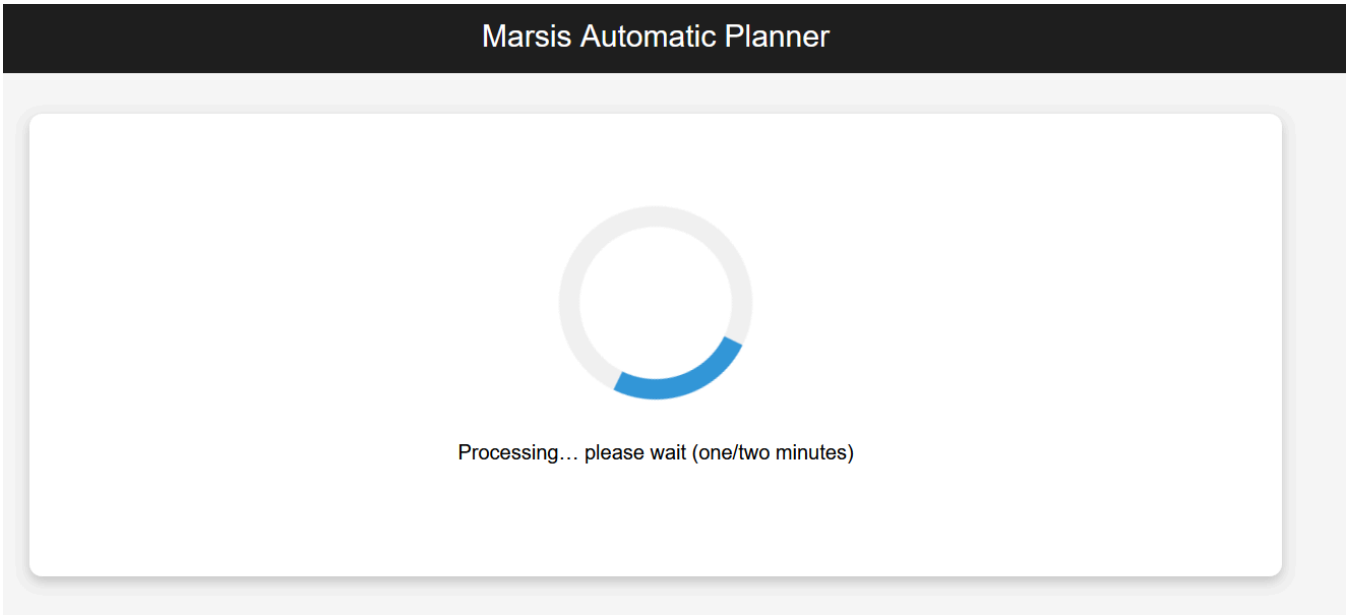

Figure 3

2.3 Result page

The Result Page (Figure 4) appears when the optimization methods are executed. By pressing the blue button at the top right, the Main Page is reloaded and all results are cleared, preventing memory overflow.
The main image displays the surface of Mars, with observations plotted on it. The colours black, red, yellow, and green represent observation quality levels from 1 to 4. Depending on the observation location, one of three views may be shown: the equator, the north pole, or the south pole of Mars.
A small red circle highlights the currently selected observation, which is indicated at the bottom of the page between two arrows. The user can navigate between observations using the arrows or by selecting the observation number directly.
Three plots are provided for each observation, as agreed with the MARSIS team. These plots show all possible observations for the given orbit, with the selected one marked by a red dot. The plots represent: (1) the radial magnetic field, (2) surface roughness, and (3) the selected quality variable (as defined in Figure 2). Each plot uses three colours to indicate value ranges: green (good), yellow (average), and red (poor).
Two green buttons allow the user to download the images displayed on the screen or to export a CSV file containing all selected observations per orbit.

Figure 4

3. Conclusions

The presented GUI provides an intuitive interface with smooth interaction, playing a key role in delivering the optimization strategy. This is achieved through the following features:

- **Graphical representation** of results using multiple plots and a planetary map
- **Intuitive interaction**, enabled by a simple loading process followed by a clear results interface
- **Customization options**, with advanced configurations accessible from the Main Page while preserving overall simplicity